Report Number:	M25-00003
Report Issue Date:	27-June-2025
Project ID:	MU2500040
Client:	Victorian Fisheries Authority
Client Contact Details:	Craig Ingram
	Level 19, 1 Spring Street, Melbourne 3000
	03-8458 2432
	0437 639 358
	craig.ingram@vfa.vic.gov.au
Quote Number:	OWL-25-00003
PO Number:	NA
Date of Sample Received:	01-May-2025
Analysis:	16S Microbial Source Tracking
Test Analyst(s):	Chi-Wen Tseng
	Timothy Lim

Sample Details

Sample #	ID	Description	Sample		Sample Type
			Date	Time	
P2500212	250421-A-W	Site A Channel Farm Site 1	21/04/2025	08:15	Water
P2500213	250421-BLJ-W	Barrier Landing Jetty	21/04/2025	08:00	Water
P2500214	250421-BLM-W	Barrier Lagoon Mouth	21/04/2025	07:50	Water
P2500215	250421-B-W	Site B Flanagan Island Farm	21/04/2025	07:30	Water
P2500216	250421-MPJ-W	Metung Pub Jetty	21/04/2025	07:20	Water
P2500217	250421-TS-W	Tambo Swamp Mouth	21/04/2025	07:10	Water
P2500218	250421-TR-W	Tambo River Mouth	21/04/2025	07:00	Water
P2500219	250421-PJ-W	Paynesville Slip Jetty	21/04/2025	06:44	Water
P2500220	250421-NAJ-W	North Arm Jetty	21/04/2025	08:30	Water
P2500221	250421-CAPO-W	Post Office Jetty, Cunningham Arm	21/04/2025	09:00	Water

Results

Table 1 - Human-specific contribution within total microbial community:

Sources	Sample #									
	P2500212	P2500213	P2500214	P2500215	P2500216	P2500217	P2500218	P2500219	P2500220	P2500221
Human	-	-	-	-	-	-	0.01%	-	-	0.00%
Unknown	100.00%	100.00%	100.00%	100.00%	100.00%	100.00%	99.99%	100.00%	100.00%	100.00%

Note for interpretation of microbial community contribution report(s):

1. Values displayed represent significant relative median microbial community contribution (relative standard deviation <100%). The results were taken as the average of five replicate SourceTracker runs.
2. Values with relative standard deviation ≥100% were deemed non-significant result, and therefore omitted from the report.
3. Values were rounded off to the nearest 2 decimal places, including those labelled as 0.00% (where applicable).
4. Unknown represents the proportion of microbial community which do not belong to any of the sources (i.e., microbial communities not of faecal-origin, or undetected sources).

Table 2 - Faecal-specific contribution within total microbial community:

Sources	Sample #									
	P2500212	P2500213	P2500214	P2500215	P2500216	P2500217	P2500218	P2500219	P2500220	P2500221
Bat	-	-	-	-	-	-	-	-	-	-
Bird	-	-	-	0.00%	-	-	-	-	-	0.00%
Cat	-	-	-	-	-	-	-	-	-	0.00%
Chicken	-	0.00%	-	-	-	-	-	-	-	0.01%
Cow	-	0.01%	-	-	-	-	0.00%	-	-	-
Deer	-	-	-	-	-	-	0.00%	-	-	-
Dog	0.00%	0.00%	-	-	-	-	-	-	-	0.00%
Fox	-	-	-	-	-	-	-	-	-	-
Horse	-	-	-	-	-	-	-	-	-	-
Human	-	-	-	-	-	-	0.01%	-	-	0.00%
Kangaroo	-	-	-	-	-	-	-	-	-	-
Possum	-	-	-	-	-	-	-	-	-	-
Rabbit	-	-	-	-	-	-	-	-	-	0.00%
Rat	-	-	-	-	-	-	0.00%	-	-	-
Sheep	-	-	-	-	-	-	-	-	-	-
Wallaby	-	-	-	-	-	-	-	-	-	-
Waterbird	-	0.00%	-	0.01%	-	-	0.00%	-	-	-
Wombat	-	-	-	-	-	-	-	-	-	-
Unknown	100.00%	99.99%	100.00%	99.99%	100.00%	100.00%	99.98%	100.00%	100.00%	99.98%

Note for interpretation of microbial community contribution report(s):

1. Values displayed represent significant relative median microbial community contribution (relative standard deviation <100%). The results were taken as the average of five replicate SourceTracker runs.
2. Values with relative standard deviation ≥100% were deemed non-significant result, and therefore omitted from the report.
3. Values were rounded off to the nearest 2 decimal places, including those labelled as 0.00% (where applicable).
4. Unknown represents the proportion of microbial community which do not belong to any of the sources (i.e., microbial communities not of faecal-origin, or undetected sources).

Table 3 - Faecal-source contribution within faecal component of microbial community:

Sources	Sample #									
	P2500212	P2500213	P2500214	P2500215	P2500216	P2500217	P2500218	P2500219	P2500220	P2500221
Bat	-	-	-	-	-	-	-	-	-	-
Bird	-	-	-	20.00%	-	-	-	-	-	18.18%
Cat	-	-	-	-	-	-	-	-	-	12.90%
Chicken	-	20.00%	-	-	-	-	-	-	-	41.38%
Cow	-	33.33%	-	-	-	-	5.13%	-	-	-
Deer	-	-	-	-	-	-	3.70%	-	-	-
Dog	66.67%	12.00%	-	-	-	-	-	-	-	6.90%
Fox	-	-	-	-	-	-	-	-	-	-
Horse	-	-	-	-	-	-	-	-	-	-
Human	-	-	-	-	-	-	51.28%	-	-	3.23%
Kangaroo	-	-	-	-	-	-	-	-	-	-
Possum	-	-	-	-	-	-	-	-	-	-
Rabbit	-	-	-	-	-	-	-	-	-	13.79%
Rat	-	-	-	-	-	-	14.81%	-	-	-
Sheep	-	-	-	-	-	-	-	-	-	-
Wallaby	-	-	-	-	-	-	-	-	-	-
Waterbird	-	27.27%	-	72.73%	-	-	20.51%	-	-	-
Wombat	-	-	-	-	-	-	-	-	-	-

Note for faecal community contribution report(s):

1. Values displayed represent significant relative median faecal community contribution (relative standard deviation <100%) . The results were taken as the average of five replicate SourceTracker runs.
2. Values with relative standard deviation ≥100% were deemed non-significant result, and therefore omitted from the report.
3. Values were rounded-off to the nearest 2 decimal places, including those labelled as 0.00% (where applicable).
4. Faecal community contribution represents normalised microbial community contribution from faecal-origin sources. The faecal community contribution from each sources was calculated by taking the median microbial community contribution of the source of interest, divided by the sum of all the microbial community contributions, excluding the microbial community contribution from unknown sources. An illustration of this is provided below, using sample P2500221 as an example.

How to interpret results:

MST results for sample P2500221

Microbial community contribution

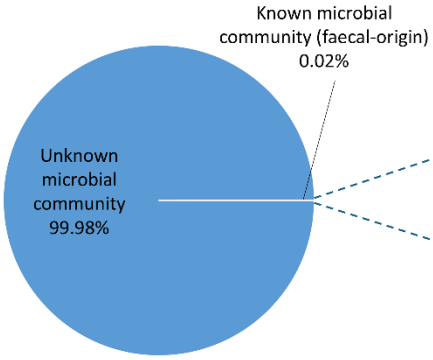

Table 2 Results

Faecal community contribution

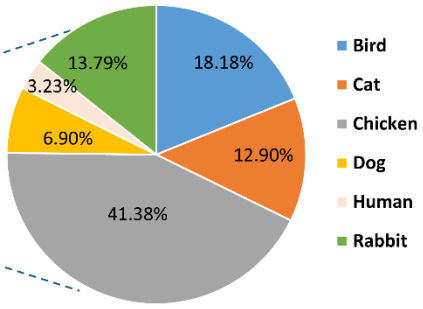

Table 3 Results

~~ End of Report ~~